

**AMENDMENT NO. 1 JANUARY 2009
TO
IS 11991 : 1986 RECOMMENDED PRACTICE
FOR FLASH BUTT WELDING OF TUBES RODS AND OTHER SECTIONS IN
CARBON
AND ALLOY STEELS**

(Page 6, clause 2.1.16, Title) — Substitute ‘Voltage’ for ‘Volt’.

(Page 6, clause 2.1.23) — Substitute ‘bonding’ for ‘bending’.

(Page 6, clause 2.1.24, line 2) — Substitute the ‘and’ for end’.

[Page 11, clause 13.1(a)] — Add the word ‘profile’ at the end of the sentence.

[Page 11, clause 13.1(d)] — Substitute ‘Irregular flash profile’ for ‘Spatter’.

(Page 15, clause 15.3) — Add the following new clause:

15.4 Process Monitoring – This is applicable only for critical welding application such as boiler tubes, etc. The machine shall be equipped with a suitable device which monitors the critical parameters for every weld. The monitoring records shall be compared with those from the procedure approval tests and those from test welds taken during production in order to verify that the same welding parameters have been employed throughout. A typical process parameter recording is shown in the Fig. 4. The records of the monitoring parameters shall be maintained at least for 5 years.

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(Pages 19 and 20, Table 2, col 9) — Substitute the following for the existing column:

FLASH TIME seconds	FLASH TIME seconds
1.00	16.0
1.50	17.0
2.00	18.0
2.50	20.0
3.25	22.0
4.00	24.0
5.00	27.0
6.00	30.0
7.00	33.0
8.00	36.0
9.00	42.0
10.0	49.0
11.0	57.0
12.0	66.0
13.0	77.0
14.0	92.0
15.0	

FIG. 4 GENERALIZED DIAGRAM ILLUSTRATING MONITORING VARIABLES